MODULE 21 · DOMAIN 4 - EVALUATION, TESTING AND OPTIMISATION

Optimising Cost and Latency, Monitoring in Production

Find the dominant cost driver, apply the levers in the right order, and run production monitoring from SLOs, alerts and runbooks.

DOMAIN 4 of 7 EXAM WEIGHT 16% MODULE 21 of 38

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Find the mortgage before cancelling the streaming subscriptions.

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Find the dominant cost driver

- **Cost is volume times tokens times price,** per model, per step.
- **Break it down:** by workflow, step, input versus output, cached versus not.
- **Sources:** usage fields on every response; the Usage and Cost Admin API by model and key.
- **Usual suspects:** a large uncached prefix, oversized retrieval, the wrong tier, long outputs, agent loops.
- **Measure cost per successful outcome,** not just per request.

> ON THE EXAM
>
> The biggest line on the bill is rarely where the team guesses. Measure before you cut.

The optimisation order

01
Measure
Find the dominant driver.

02
Zero-risk first
Cache static prefixes; drop waste.

03
Fewer tokens
Trim retrieval and output, if evals hold.

04
Route
Lighter tiers for tested easy cases.

05
Batch
Where latency allows and policy permits.

06
Re-measure
Cost and quality, every time.

ON THE EXAM

Moves that cannot change an answer go first. Moves that might, need an evaluation result.

Cost levers, October 2026

LEVER	WHAT IT SAVES	WATCH OUT FOR
Prompt caching	Reads 0.1x input; 0.05x on Opus, Sonnet 5.5	Writes 1.25x or 2x; stable prefix
Message Batches	50% off; most within an hour	Up to 24 h; not ZDR or HIPAA
Routing	Haiku input: a fortieth of Opus	Only where evals hold
Shorter output	Output is 5x the input price	Thinking is billed as output
Fewer chunks	Input on every request	Test recall first

ON THE EXAM

Batch is not offered on Bedrock, Google Cloud or Foundry. Check the platform before planning on it.

Capacity and rate limits

- **Limits are per model class:** requests, input tokens and output tokens per minute.
- **A 429 carries retry-after.** Back off and retry; do not hammer.
- **Cached reads do not count** towards the input-token limit: caching buys headroom.
- **Routing spreads load** across model classes, each with its own limits.
- **Plan for peaks,** not averages: reminder surges, month-end, incidents.

> ON THE EXAM
>
> Capacity is a cost and latency concern too. Throttled requests are slow requests.

Production monitoring, in order

01
Define SLOs
What users are promised.

02
Error budget
How much failure is allowed.

03
Instrument
Only what the SLOs need, plus traces.

04
Dashboards
SLOs, cost, quality samples.

05
Alerts
On budget burn, not every blip.

06
Runbooks
The first steps for each alert.

ON THE EXAM

SLOs and error budgets come before instrumentation. Otherwise you measure everything and watch nothing.

What to alert on

SIGNAL	ALERT WHEN	RUNBOOK STARTS WITH
Latency or availability SLO	Error budget burns fast	Recent changes; traces of slow steps
Sampled quality score	Below the floor for a set window	Versions, retrieval, failing slices
Cost per outcome, cache hit rate	Departs from baseline	What changed in the prefix or mix
Errors, timeouts, 429s	Rate above threshold	Capacity, limits, provider status
Safety and policy flags	Any confirmed breach	Contain, review, escalate

ON THE EXAM

An alert nobody can act on is noise. Every alert needs an owner and a first step.

Drift without a deployment

- **The input mix shifts:** longer documents, new topics, a new customer segment.
- **A small change breaks the cache,** and cost jumps silently.
- **Prompts and tool lists grow,** one well-meant addition at a time.
- **Agent loops lengthen** as tasks get harder.
- **Model retirements** force changes on a fixed date.

> ON THE EXAM
>
> Watch cost and quality per workflow against a baseline. Drift is only visible as a trend.

> Mereside asks Corvane to cut agent-assist running costs by 30%. The team proposes moving all traffic to the lightest tier.
>
> Usage data shows 65% of input tokens are retrieved policy chunks: 40 per question. The evaluation shows accuracy unchanged at 12.

What should Corvane recommend first?

A Move all traffic to the lightest tier, as proposed

B Cut retrieved chunks to the level the evaluation supports, then re-measure

C Ask Claude for shorter answers

D Move agent assist to batch processing

◆ PAUSE AND DECIDE

What should Corvane recommend first?

A Move all traffic to the lightest tier, as proposed

An untested quality trade, aimed away from the driver.

B Cut retrieved chunks to the level the evaluation supports, then re-measure

Hits the dominant driver, with quality already shown to hold.

C Ask Claude for shorter answers

Output is not where the money goes here.

D Move agent assist to batch processing

Live calls cannot wait hours; and it is on Bedrock.

THE SIGNAL

Find the dominant driver, then take the cut the evaluation already proves is safe.

> Quillon re-summarises about 30,000 discharge records each night for a quality audit. The work runs in its HIPAA-ready API organisation.
>
> Finance asks Corvane to move it to the Message Batches API, for the 50% discount.

What should Corvane recommend?

A. Use the Message Batches API, since nobody needs the results quickly

B. Use batches after removing patient names from each record

C. Run it through batches in a separate, non-HIPAA organisation

D. Keep it on eligible features, and cut cost with caching, routing and shorter outputs

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Use the Message Batches API, since nobody needs the results quickly

Batch is excluded from HIPAA-ready use. Latency is not the issue.

B. Use batches after removing patient names from each record

Removing names is not de-identification.

C. Run it through batches in a separate, non-HIPAA organisation

Patient data outside BAA coverage.

D. Keep it on eligible features, and cut cost with caching, routing and shorter outputs

Hard constraint first; then optimise inside it.

THE SIGNAL

A discount that breaks a compliance boundary is not an option. Optimise inside the boundary.

> Tallyhawk is launching its redesigned support agent next month and wants production monitoring in place first.
>
> An engineer proposes instrumenting every available metric and building dashboards for all of them.

What should come first?

A. Build dashboards for every available metric, then decide what matters

B. Agree SLOs and error budgets with the business, then instrument what they need

C. Alert on every error from day one

D. Buy an observability product and use its default dashboards

◆ PAUSE AND DECIDE

What should come first?

A Build dashboards for every available metric, then decide what matters

Measuring everything, watching nothing.

B Agree SLOs and error budgets with the business, then instrument what they need

Monitoring starts from what users are promised.

C Alert on every error from day one

Alert fatigue within a week.

D Buy an observability product and use its default dashboards

A tool, not a decision about what matters.

THE SIGNAL

SLOs and error budgets before instrumentation. Decide what matters, then measure it.

> Quillon's portal pages the on-call engineer for every single request slower than 2 seconds. It fires about 40 times a night.
>
> The engineers have started muting it.

What should Corvane recommend?

A Raise the threshold so only very slow requests page

B Remove latency alerting and rely on the dashboard

C Send the alerts by email instead of paging

D Alert on error-budget burn: page on fast burn, ticket on slow, each with a runbook

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Raise the threshold so only very slow requests page

Still per-request noise, now blind to the real target.

B Remove latency alerting and rely on the dashboard

Nobody watches a dashboard at three in the morning.

C Send the alerts by email instead of paging

Moves the noise to an inbox.

D Alert on error-budget burn: page on fast burn, ticket on slow, each with a runbook

Alerts tied to the promise, sized to the urgency.

THE SIGNAL

Alert on the SLO, not on individual requests. A muted alert is worse than none.

> Tallyhawk's cost per resolved conversation rose 40% over two months. No model or prompt release happened in that time.
>
> The dashboard shows the cache hit rate fell from 85% to 20% three weeks ago.

What should Corvane investigate first?

A What started varying in the cached prefix three weeks ago, to restore a stable prefix

B Moving all conversations to the lightest tier

C Capping every conversation at a fixed number of turns

D Negotiating a volume discount with Anthropic

◆ PAUSE AND DECIDE

What should Corvane investigate first?

A What started varying in the cached prefix three weeks ago, to restore a stable prefix

The monitoring already points to the cause and the date.

B Moving all conversations to the lightest tier

A quality trade that ignores the evidence.

C Capping every conversation at a fixed number of turns

Hurts customers; the cause is the cache.

D Negotiating a volume discount with Anthropic

Discounts a self-inflicted cost.

THE SIGNAL

Cost drift with a cache-hit collapse: something in the prefix started changing. Find it.

> Tallyhawk wants to cut Claude costs without any risk to answer quality.
>
> Its support agent resends a long, unchanging system prompt and tool list. A nightly job summarises new help articles; nobody needs it before morning, and it holds no customer data.

Which two moves cut cost with no risk to quality? Choose two.

- A Move all support traffic to the lightest tier
- B Halve the output limit on every request
- C Cache the stable system prompt and tool definitions
- D Lower effort on every request
- E Run the nightly summarisation job through the Message Batches API

◆ PAUSE AND DECIDE

Which two moves cut cost with no risk to quality? Choose two.

A Move all support traffic to the lightest tier

A quality trade on hard cases.

B Halve the output limit on every request

Risks truncated answers.

C Cache the stable system prompt and tool definitions

Same answers, cheaper reads.

D Lower effort on every request

Less reasoning where it is needed.

E Run the nightly summarisation job through the Message Batches API

Same model, same output, half price; timing fits.

THE SIGNAL

No risk to quality means caching and batching: same model, same output, lower price.

In five lines

- **Find the dominant driver first:** by workflow, step, input, output and cache.
- **Order:** zero-risk caching, then fewer tokens, routing, and Batch where allowed.
- **Constraints first:** the Batch API is excluded from HIPAA-ready use and not offered on Bedrock.
- **Monitoring:** SLOs and error budgets, then instruments, dashboards, alerts, runbooks.
- **Alert on budget burn and drift,** not on every slow request.

COMING UP

Domain 4 complete. Next, Domain 5: guardrails and LLM failure modes.